

TECHNICAL PROJECT PORTFOLIO:

PROJECT BETA

Systems Engineer: Satnam Singh

Domain Stack: React, Node.js, Express API, PostgreSQL

Target Tracking: IT Support Engineering / Infrastructure Analysis

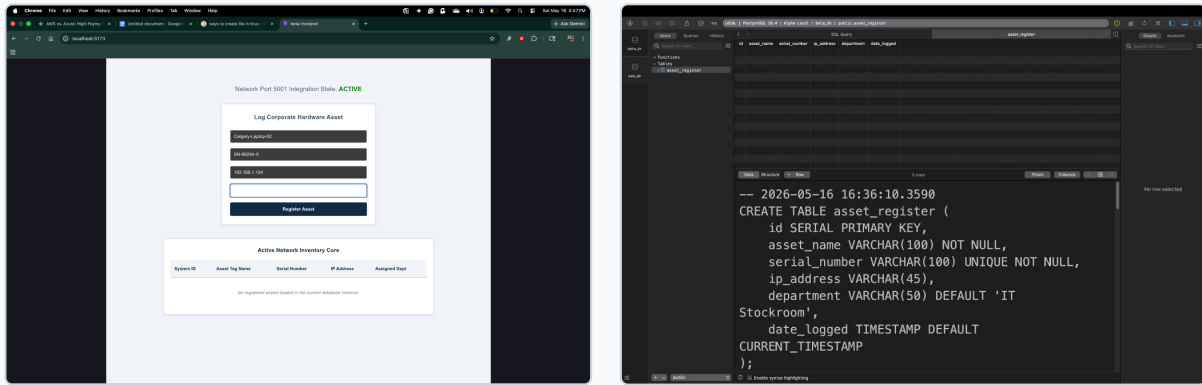
1. EXECUTIVE SUMMARY & SYSTEM OBJECTIVE

In modern enterprise networks, keeping accurate, real-time tracking profiles of deployment metrics across shifting operational tiers is a primary administrative dependency. Relying on manual inventories introduces data duplication and systemic network blind spots. **Project Beta** mitigates these issues by building a decoupled three-tier system loops designed to safely register hardware profiles and verify operational connections seamlessly.

2. RELATIONAL SCHEMA STRUCTURE VALIDATION

Data boundaries are strictly enforced inside the database cluster layer. Unique validation rules block asset duplicate errors from reaching log files during deployment operations:

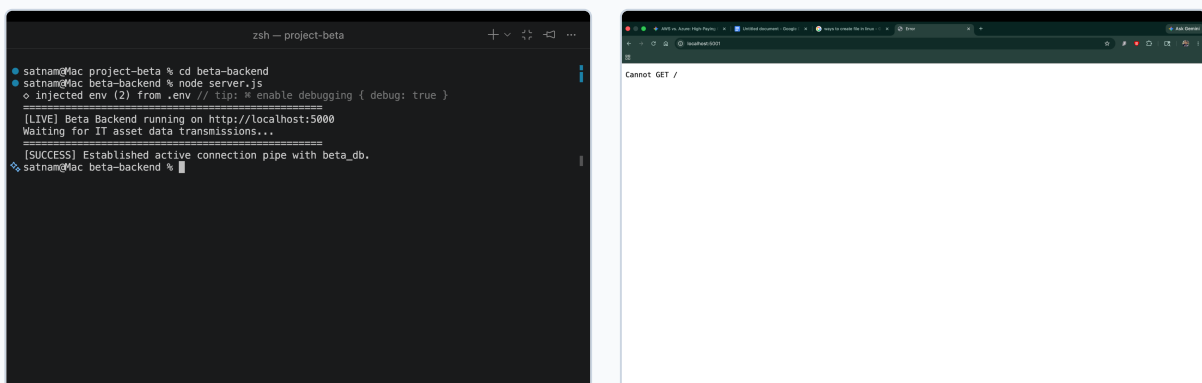
```
CREATE TABLE asset_register (  
  id SERIAL PRIMARY KEY,  
  asset_name VARCHAR(100) NOT NULL,  
  serial_number VARCHAR(100) UNIQUE NOT NULL,  
  ip_address VARCHAR(45),  
  department VARCHAR(50) DEFAULT 'IT Stockroom',  
  date_logged TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```



Verification Proof 1 & 2: Relational Table Configuration Matrix and Column Constraints inside TablePlus.

3. INFRASTRUCTURE TRIAGE & PORT RESOURCE MANAGEMENT

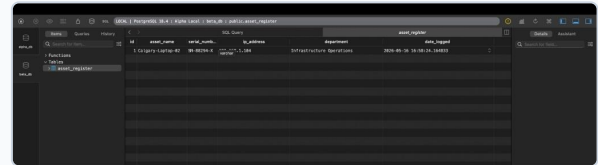
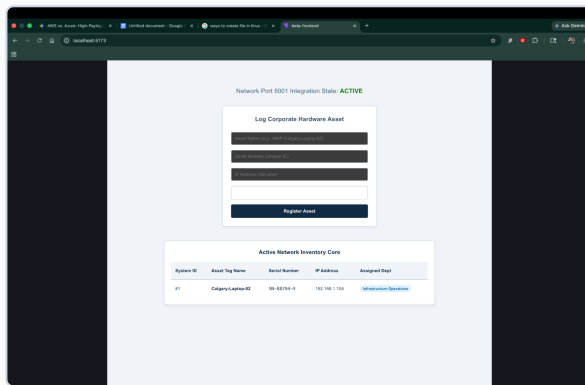
Localized system process tracking caught an endpoint port resource overlap. Background streaming listeners on macOS default to grabbing network Port 5000, causing connection blockages. To fix the line, application routes were decoupled into environment variables and the live API communication channel was securely migrated to **TCP Port 5001**.



Verification Proof 3 & 4: Live Environment Handshake and Operational Node.js API Pipeline Listening Status.

4. ASYNCHRONOUS CLIENT INTERFACE LOOPS

The frontend portal utilizes state management hooks to capture parameters cleanly. When an entry is saved, the application drives a background asynchronous network payload to the API. On database success, a verification retrieval loops back to automatically draw the updated records onto the core dashboard matrix without requiring viewport reloads.



Verification Proof 5 & 6: Stateful Frontend Console and Real-time Relational Data Ingestion Grid Views.